Exercise Sheet 1

1. A sequence $\{X_t | t \in \mathbb{Z}\}$ is called independent, if each finite subcollection is independent, i.e. if

$$P(X_{t_1} \leq x_1 \dots X_{t_n} \leq x_n) = \prod_{i=1}^{n} P(X_{t_i} \leq x_i).$$

   Let $\{F_t | t \in \mathbb{Z}\}$ be a given sequence of distribution functions. Use Kolmogorov's extension theorem to prove the existence of an independent sequence $\{X_t | t \in \mathbb{Z}\}$ with given marginal distributions $F_t(x)$.

2. Let $\{X_t | t \geq 0\}$ be the random walk model $\{\varepsilon_t\} \sim \text{IID}(0, 1)$.

$$X_t = X_{t-1} + \varepsilon_t, \quad t \geq 1, \quad X_0 = 0.$$

   Calculate the mean- and the auto-covariance function.

3. Assume X_i have a finite variance and $a_i \in \mathbb{R}$, $1 \leq i \leq n$. We have used in the lecture that

$$\text{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i,j=1}^{n} a_i a_j \text{Cov}(X_i, X_j).$$

   Verify this.

4. Let $\{Z_t | t \in \mathbb{Z}\} \sim \text{IID}(0, 1)$. Let a, b and c be constants. Which, if any, of the processes are stationary?

   - $X_t = a + bZ_t + cZ_{t-1}$;
   - $X_t = Z_1 \cos(ct) + Z_2 \sin(ct)$;
   - $X_t = Z_t \cos(ct) + Z_{t-1} \sin(ct)$;

5. Same as in the last exercise. Which, if any, of the processes are stationary?

   - $X_t = a + bZ_0$;
   - $X_t = Z_0 \cos(ct)$;
   - $X_t = Z_t Z_{t-1}$.

6. Let Θ be uniformly distributed on $[0, 2\pi)$ and let Z be independent of Θ have a standard exponential distribution [the density of Z is given by $f_Z(t) = e^{-t}$ if $t > 0$ and 0 else]. Compute the ACF of

$$X_t = Z \cos(t + \Theta).$$

7. Which, if any, of the following functions defined on the integers is the ACF of a stationary time series?

- $f(h) = 1$ if $h = 0$ and $f(h) = 1/h$ if $h \neq 0$.
- $f(h) = (-1)^{|h|}$.

8. Which, if any, of the following functions defined on the integers is the ACF of a stationary time series?

   - $f(h) = 1$ if $h = 0$, $f(h) = 0.4$ if $h = \pm 1$ and $f(h) = 0$ otherwise.
   - $f(h) = 1$ if $h = 0$, $f(h) = 0.6$ if $h = \pm 1$ and $f(h) = 0$ otherwise.